Demand Response Schemes under a Single-Buyer Model for Consumers and Supplier by using the Bi-level programming

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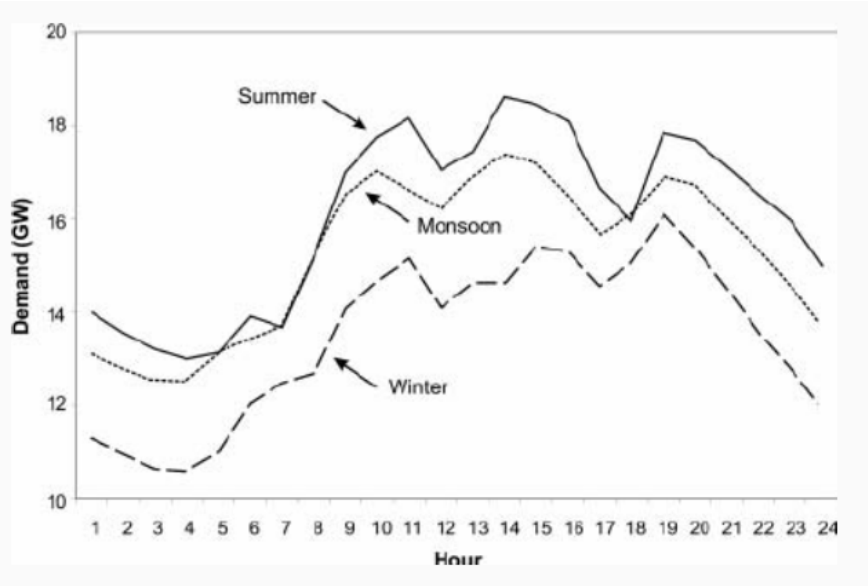
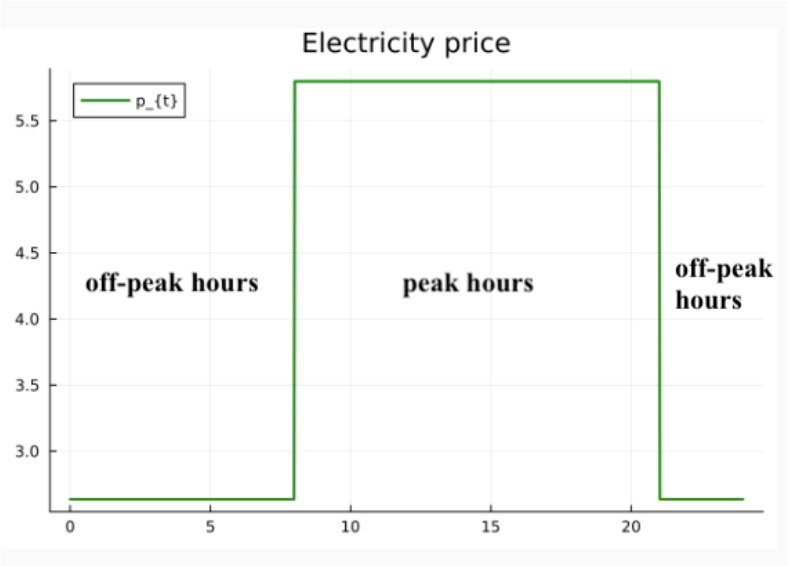
Demand Response

Demand Response is a policy that encourages consumers to shift their electricity consumption to times when the demand is lower.





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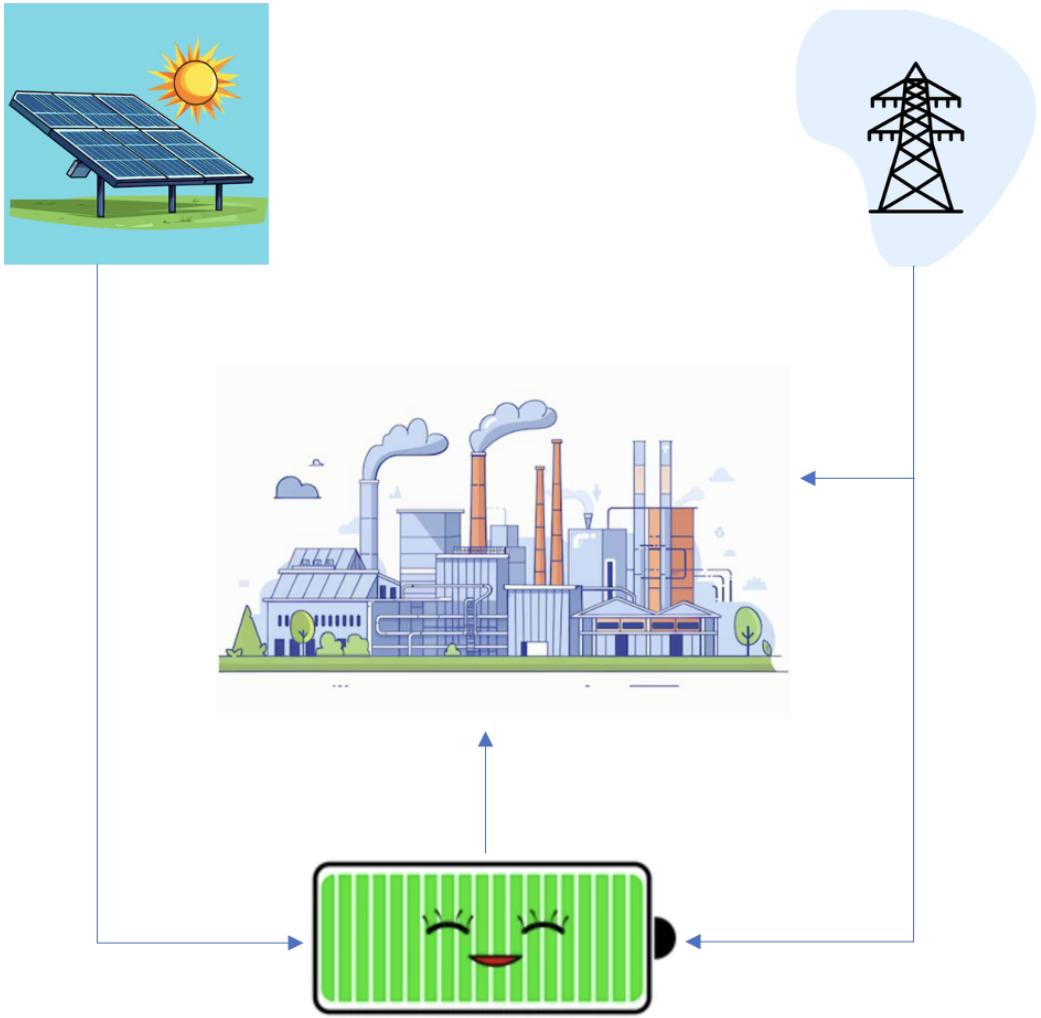
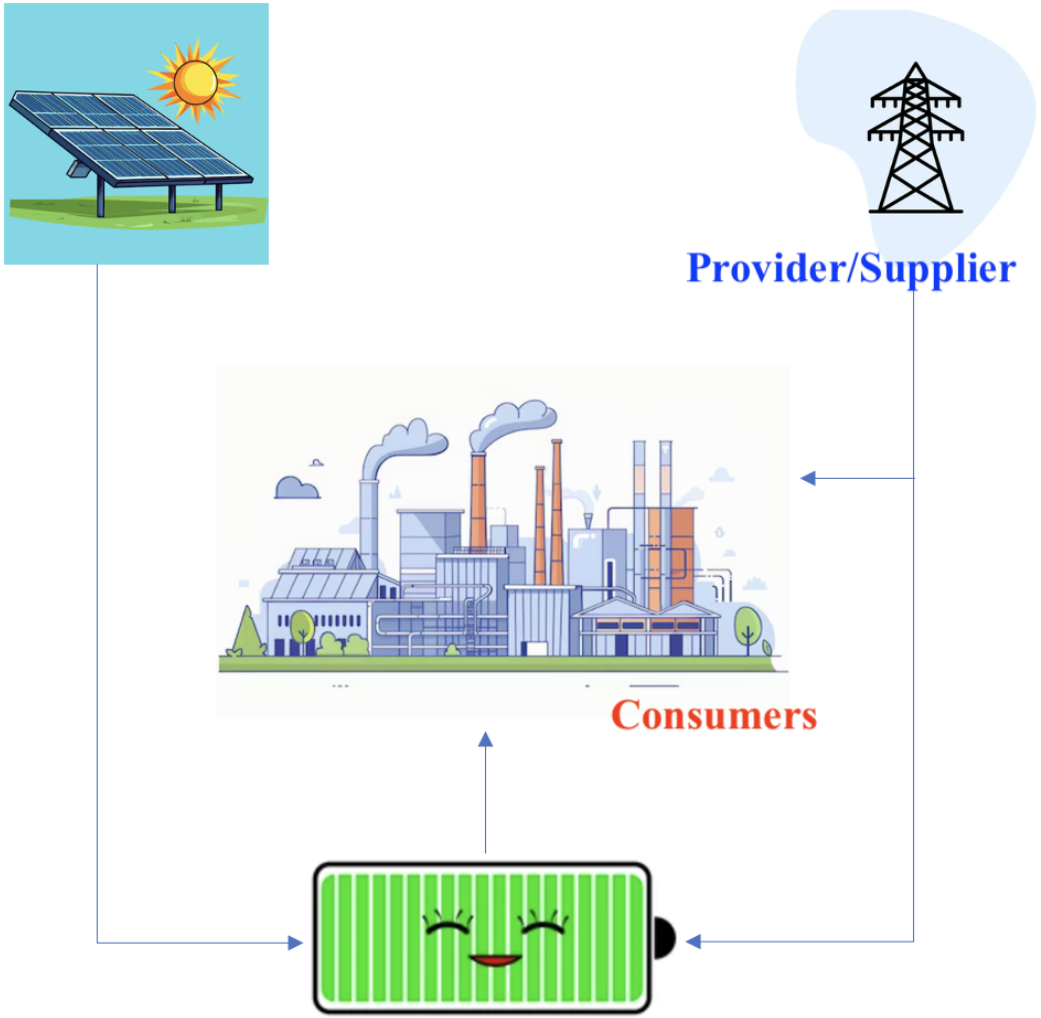
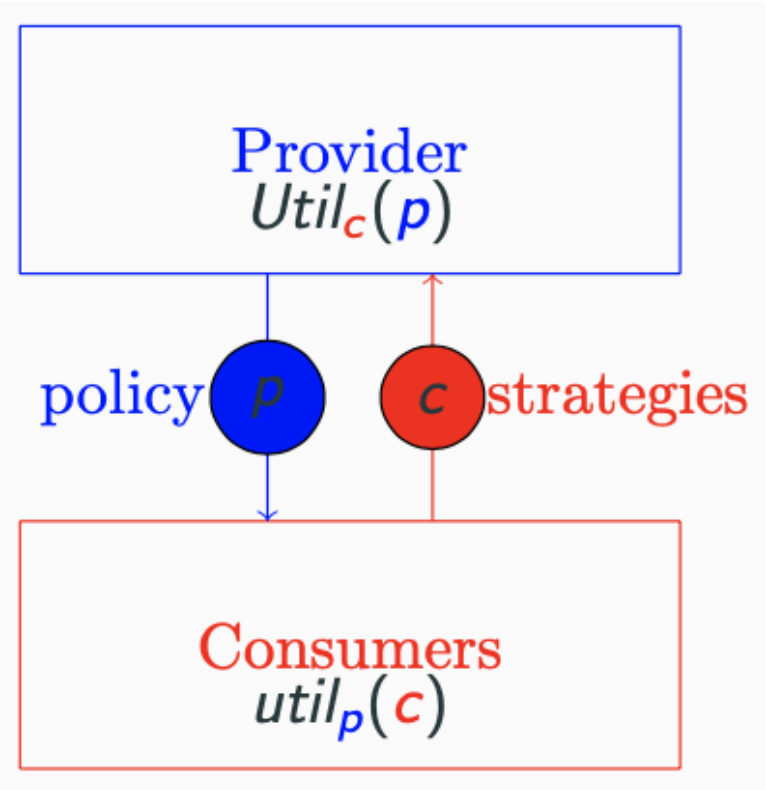
Energy System...



Energy System...



Bi-level model



The provider offers a policy .

The consumers respond with their strategies.

The utilities of both actors are affected by both policy and strategies.

In this talk, we present a model describing the interactions between the provider and the consumers.

Bi-level model

Definition

$$
\begin{aligned}
\text{"} \min_{x \in X, y} \text{"} \quad & F(x, y) && \text{(1a)} \\
s.t. \quad & G(x, y) \geq 0 && \text{(1b)} \\
& y \in S(x), && \text{(1c)}
\end{aligned}
$$

where $S(x)$ is the set of optimal solutions of the x-parameterized problem

$$
\begin{aligned}
\min_{y \in Y} \quad & f(x, y) && \text{(2a)} \\
s.t. \quad & g(x, y) \geq 0. && \text{(2b)}
\end{aligned}
$$

Model

$a \in \mathbb{A} = \{1, \ldots, A\}$	Appliances
$t \in \mathbb{T} = \{1, \ldots, 24\}$	Time of the day
c_t	Quadratic coefficient for the generation cost
ℓ	Transmission capacity of electricity from the grid
P	Maximum tariff per unit
d_a	Daily demand for an appliance a
κ	Capacity of the storage
ρ_t	Charging rate of the storage
σ	The energy generated by solar PV
$q_{a,t}$	Monetary inconvenience cost to operate the appliance a at time t

Model

p_t	The electricity price.
$e_t^{grid \to a}$	Decision variable describing the flow from the grid to the appliance a at time t.
$e_t^{sto \to a}$	Decision variable describing flow from the storage to the appliance a at time t.
$e_t^{grid \to sto}$	Decision variable describing flow from the grid to the storage at time t.
$e_t^{solar \to sto}$	Decision variable describing flow from the solar PV to the storage at time t.
e_t^{sto}	Variable describing the remaining electricity amount in the storage after charging at time t.

Model

For the whole day, consumers pay

$$\sum_{t\in\mathbb{T}}\sum_{a\in A}\left[\underbrace{p_t(e_t^{grid\to a}+e_t^{grid\to sto})}_{\textit{electricity bill}}+\underbrace{q_{a,t}(e_t^{grid\to a}+e_t^{sto\to a})}_{\textit{monetary inconvenience cost}}\right]$$

Model

Storage constraints:

$$e_1^{sto} = 0 \tag{3a}$$

$$e_t^{sto} = e_{t-1}^{sto} - \sum_{a \in \mathbb{A}} e_{t-1}^{sto \to a} + e_{t-1}^{grid \to sto} + e_{t-1}^{solar \to sto}, t = 2, \cdots, 24 \tag{3b}$$

$$e_t^{sto} \leq \kappa \tag{3c}$$

$$\sum_{a \in \mathbb{A}} e_t^{sto \to a} \leq e_t^{sto} \tag{3d}$$

$$e_t^{solar \to sto} \leq \sigma_t \tag{3e}$$

$$e_t^{grid \to sto} + e_t^{solar \to sto} \leq \rho \tag{3f}$$

for all $t \in \mathbb{T}$ and $a \in \mathbb{A}$.

Model

Loads constraints:

$$\sum_{t \in \mathbb{T}} (e_t^{grid \to a} + e_t^{sto \to a}) = d_a, \quad a \in \mathbb{A} \tag{4a}$$

$$\sum_{a \in \mathbb{A}} e_t^{grid \to a} \leq \ell, \quad t \in \mathbb{T} \tag{4b}$$

Model

Electricity flow constraints:

$$e_t^{grid \to a} \geq 0 \tag{5a}$$
$$e_t^{sto \to a} \geq 0 \tag{5b}$$
$$e_t^{grid \to sto} \geq 0 \tag{5c}$$
$$e_t^{solar \to sto} \geq 0 \tag{5d}$$
$$e_t^{sto} \geq 0 \tag{5e}$$

for all $t \in \mathbb{T}$ and $a \in \mathbb{A}$.

Model

Let the decision variable $(e_t)_{t\in\mathbb{T}}$ such that

$$e_t = (e_t^{grid\to a}, e_t^{sto\to a}, e_t^{grid\to sto}, e_t^{solar\to sto}, e_t^{sto})_{a\in\mathbb{A}}.$$

The follower problem

$$\min_{e_t} \quad \sum_{t\in\mathbb{T}}\sum_{a\in A}\left[p_t(e_t^{grid\to a} + e_t^{grid\to sto}) + q_{a,t}(e_t^{grid\to a} + e_t^{sto\to a})\right] \tag{6a}$$

$$s.t. \quad (3), (4), (5) \tag{6b}$$

Model

For the whole day, the supplier obtains the marginal profit

$$\sum_{t \in \mathbb{T}} \sum_{a \in \mathbb{A}} (p_t - c_t)(e_t^{grid \to a} + e_t^{grid \to sto}) \tag{7}$$

Model

The day-ahead problem is a bi-level programming of

$$\max_{p_t, e_t} \quad \sum_{t \in \mathbb{T}} \sum_{a \in \mathbb{A}} (p_t - c_t)(e_t^{grid \to a} + e_t^{grid \to sto}) \tag{8a}$$

$$s.t. \quad p_t \in [0, P] \tag{8b}$$

$$e_t \in \Psi(p_t), \tag{8c}$$

where $\Psi(p_t)$ is the solution set mapping of the follower problem (6).

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Methodology

Q: How can we solve the bi-level programming?

A: Transform into a single-level programming.

Q: How can we transform the bi-level into a single-level programming?

A: We use **KKT conditions!**

Methodology

Luckily ...

The follower problem is **linear** then it is **"convex"**, so **KKT conditions!**

Methodology

1. KKT reformulation to the follower problem $\Rightarrow$ *MPCC*

$$\max_{p_t, e_t} \quad \sum_{t \in \mathbb{T}} \sum_{a \in \mathbb{A}} (p_t - c_t)(e_t^{grid \to a} + e_t^{grid \to sto}) \tag{9a}$$

$$s.t. \quad p_t \in [0, P] \tag{9b}$$

$$(e_t, \lambda_t, \nu_t) \in \Phi(p_t), \text{ for } \quad t \in \mathbb{T} \tag{9c}$$

$$\Downarrow$$

QPCC

Methodology

The difficulty is **complementarity slackness.**

$$\nu_t^{(1)}(e_t^{sto} - \kappa) = 0 \tag{10a}$$

$$\nu_t^{(2)}\left(\sum_{a\in\mathbb{A}} e_t^{sto\to a} - e_t^{sto}\right) = 0 \tag{10b}$$

$$\nu_t^{(3)}(e_t^{solar\to sto} - \sigma_t) = 0 \tag{10c}$$

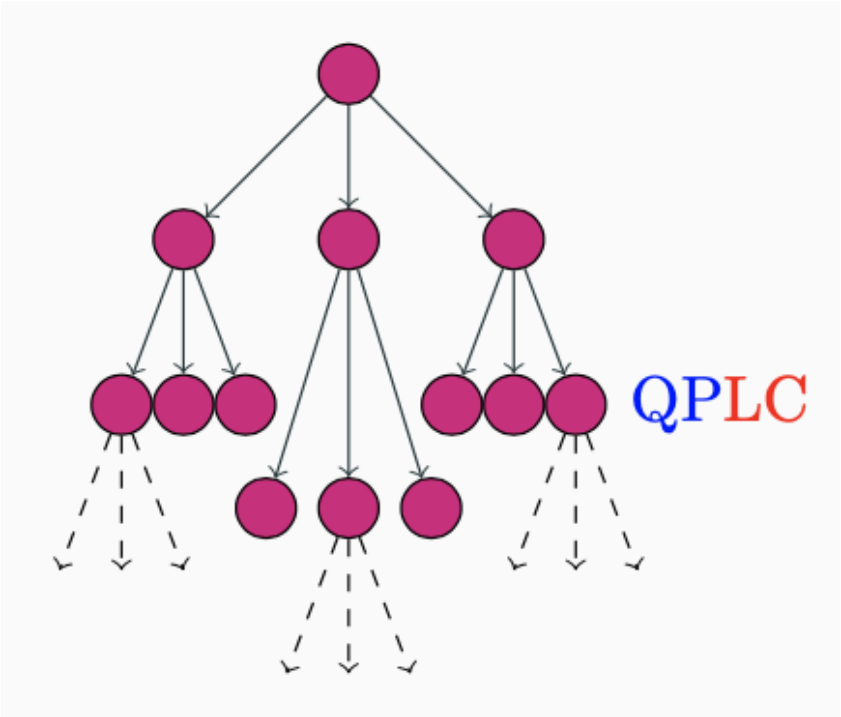
$$\nu_t^{(4)}(e_t^{grid\to sto} + e_t^{solar\to sto} - \rho) = 0 \tag{10d}$$

$$\nu_t^{sto} \cdot e_t^{sto} = 0 \tag{10e}$$

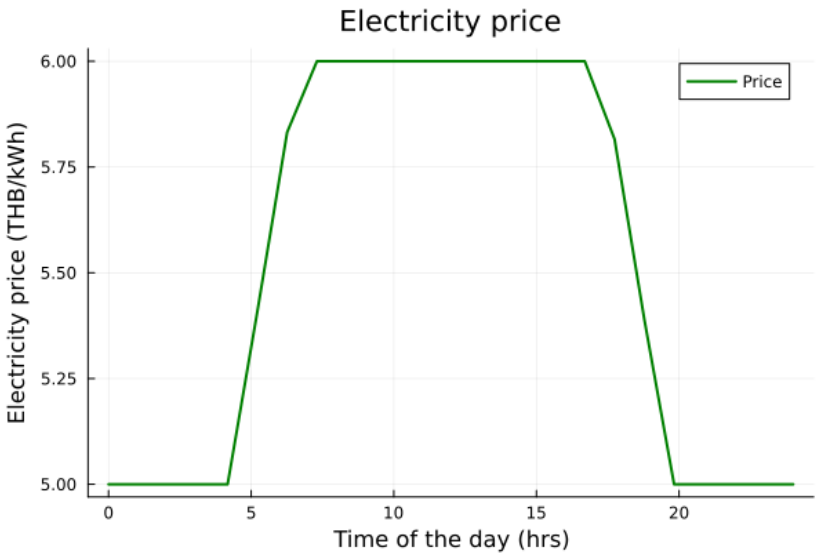
They are **non-convex!**

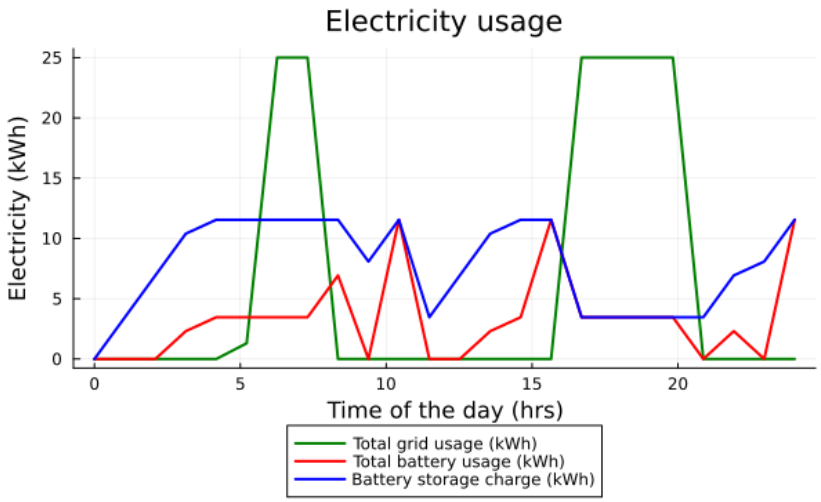
Methodology

2. Applying SOS1 approach to complementarity slackness.

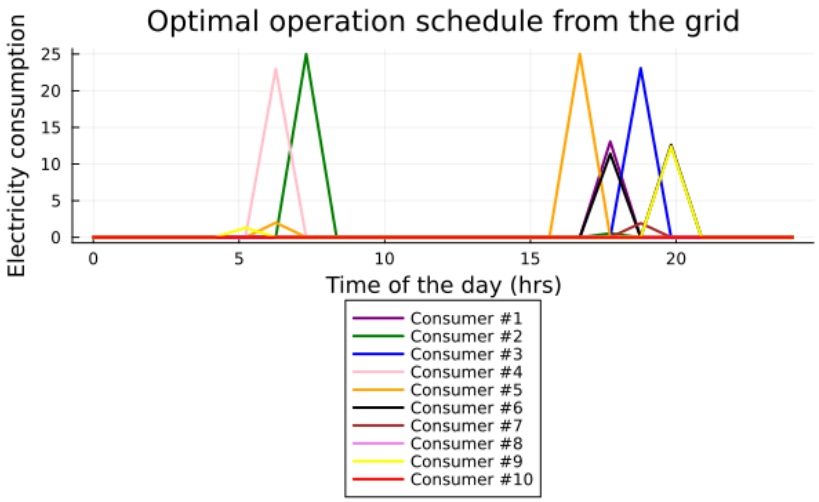


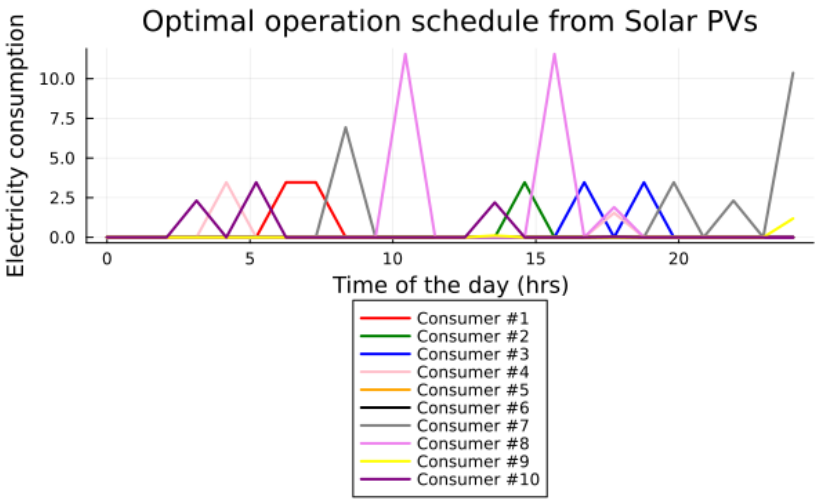
Results


Electricity price
Price
Electricity price (THB/kWh)
6.00
5.75
5.50
5.25
5.00
0
5
10
15
20
Time of the day (hrs)


Electricity usage
Electricity (kWh)
25
20
15
10
5
0
0
5
10
15
20
Time of the day (hrs)
Total grid usage (kWh)
Total battery usage (kWh)
Battery storage charge (kWh)

Results


Optimal operation schedule from the grid
Electricity consumption
Time of the day (hrs)
Consumer #1
Consumer #2
Consumer #3
Consumer #4
Consumer #5
Consumer #6
Consumer #7
Consumer #8
Consumer #9
Consumer #10


Optimal operation schedule from Solar PVs
Electricity consumption
Time of the day (hrs)
Consumer #1
Consumer #2
Consumer #3
Consumer #4
Consumer #5
Consumer #6
Consumer #7
Consumer #8
Consumer #9
Consumer #10

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Conclusion

We have used the bi-level programming in an energy system under the demand response program. The bi-level programming showed the interactions between two actors in the system perfectly.

Acknowledgement

Thanks...

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Thank you for your attention!